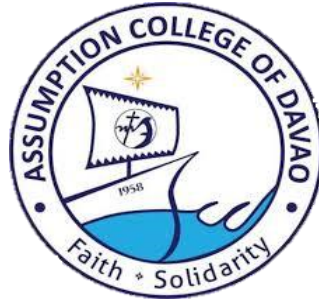


Automatic Plant Watering System



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IT 105 – DISCRETE MATHEMATICS for IT

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1. AutomatedWateringPlant

This project focuses on automating the process of watering plants by using a moisture detection system and basic electronic components.

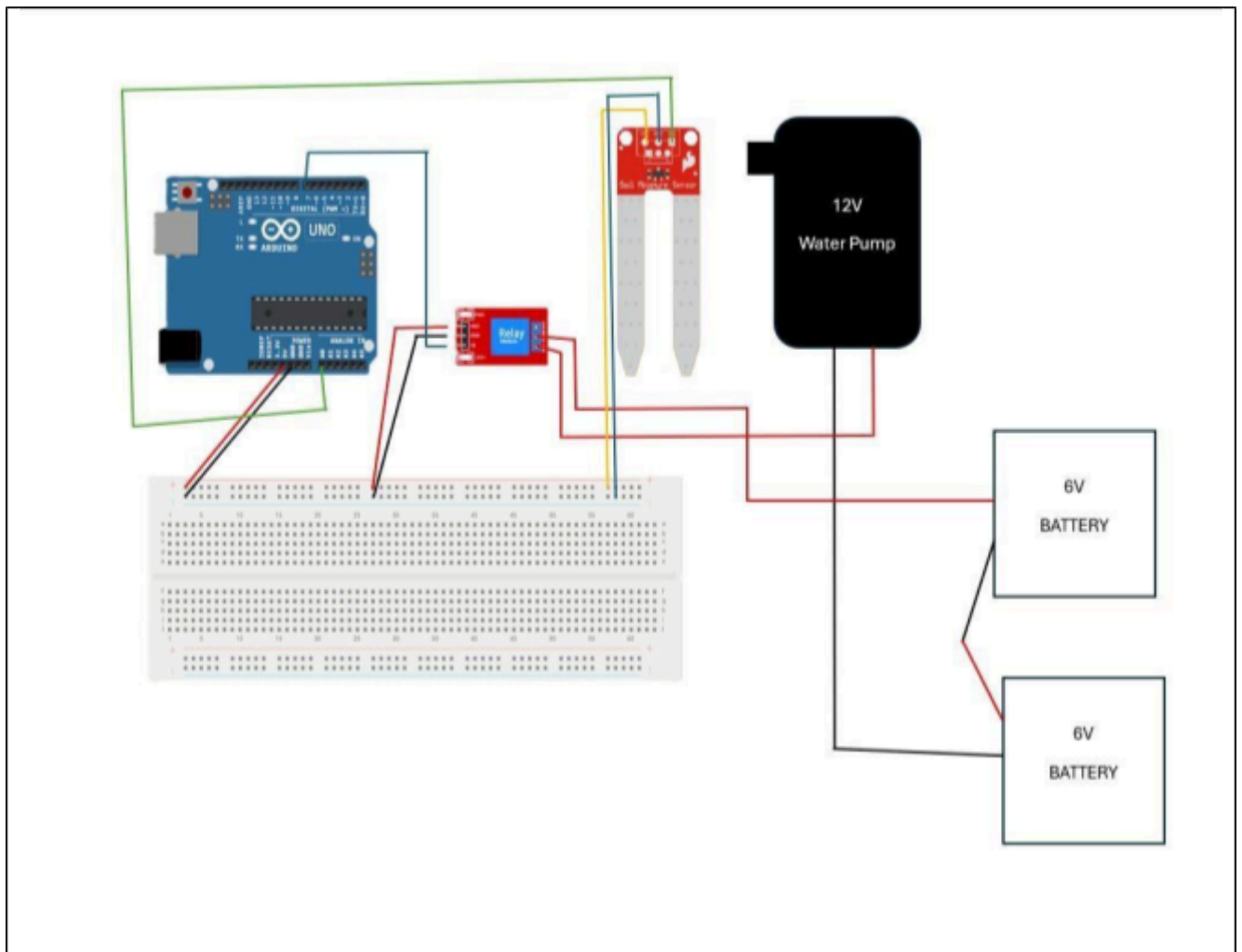
2. Objective

The objective of this project is to develop a simple automated system that can monitor soil moisture and water plants when necessary. By using an Arduino Uno, soil sensor, and relay module, the system detects whether the soil is dry or wet and activates a water pump accordingly. This helps maintain plant health and reduces the need for manual watering.

3. Components Used

Components	Quantity	Notes
Arduino Uno	1	Main Controller of the system
Jumper Wires	20	Used for Connections
Relay Module (Single Channel)	1	Controlled the Water Pump
Soil Sensor	1	Detects moisture level
Water Pump	1	Pumps water to the plant
12V Battery	1	Power source for the Water Pump to work
BreadBoard	1	Used for circuit set-up

4.Circuit



5. TruthTable

SoilCondition	WaterPump(Output)
0 (Wet)	Off
1 (Dry)	On

6. Step-by-StepInstructions

To assemble the system, begin by placing the Arduino Uno on a stable surface and setting up the breadboard. Connect the soil sensor to the Arduino using jumper wires, ensuring proper connection to VCC, GND, and signal pin. Next, connect the

relay module to the Arduino, linking its VCC, GND, and input pin correctly. After that, connect the water pump to the relay module and attach it to the 12V battery as its power source. Once all components are connected, upload the program to the Arduino Uno. Finally, insert the soil sensor into the soil and turn on the system to test its functionality.

7. Observations

During testing, the system successfully responded to soil moisture changes. When the soil was dry, the water pump automatically turned on and watered the plant. When the soil became wet, the pump stopped working. It was also observed that the system's performance depends on proper sensor placement and stable wiring connections.

8. Challenges/Notes

One of the main challenges encountered was correctly wiring the relay module and ensuring proper connections using the breadboard and jumper wires. There were also slight conflict in sensor readings, which required adjustments in positioning. Another issue was really hard to us is when we try all materials we use and we can't find first where it was wrong before we realize that the positioning of all the wires is wrong so we need to fix and start again.

9. Conclusion

The Automated Plant Watering System shows how easy it is to take everyday electronics and create a solution to an everyday problem. The use of an Arduino Uno, soil sensor, and relay module to create a system that automatically waters based on soil condition can make less manual work for someone and promotes healthy plants by watering according to their true needs. There were some minor challenges associated with setup and calibration, but overall the project was successful, and there is room for improvement on the project to produce more automated solutions.